## Inputs

### Project Description

```
This project controls servo motor with a potentiometer.
                        ⋮
When the Arduino powers up, it should:
1. Generate Servo object
2. Set potentiometer signal pin a0 as INPUT
3. Continuously read the potentiometer value, which
ranges from 0 to 1023, and map it to a range of 0 to
180
4. Write the mapped value to the servo motor
5. Continuously read the servo value and print it to
the serial monitor
```

### Code

```
void setup() {
    Serial.begin(9600);
    myservo.attach(SERVO_PIN);
    pinMode(POT_PIN, INPUT);
}
void loop() {
    int pot = analogRead(POT_PIN);
    myservo.write(map(pot, 0, 1023, 0,
180));
    Serial.println(myservo.read());
    delay(100);
}
```

## Evaluation

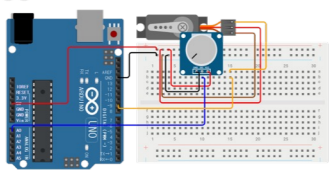
### Test Procedure

```
Check if green LED is on

Press button
→ Check if yellow LED is on

Wait 1 sec
→ Check if red LED is on

Wait 2 sec
→ Check if green LED is on
```

**Success** or **Failure**

## Generated Circuit

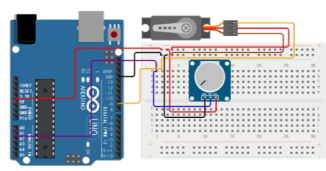
### o3-Mini



**Success**

| | |
|---|---|
| Redundant Connection | 0 |
| Extraneous Component | 0 |
| Missing Component | 0 |
| Isolated Component | 0 |
| Pin Conflict | 0 |
| Breadboard Bypass | 0 |

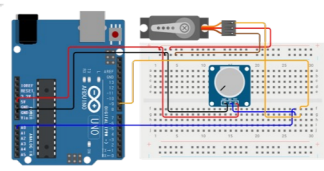
### Claude-3.7-Sonnet



**Failure**

| | |
|---|---|
| Redundant Connection | 0 |
| Extraneous Component | 6 |
| Missing Component | 0 |
| Isolated Component | 6 |
| Pin Conflict | 2 |
| Breadboard Bypass | 0 |

### Mistral-Small 3

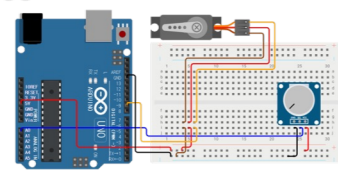


**Failure**

| | |
|---|---|
| Redundant Connection | 0 |
| Extraneous Component | 0 |
| Missing Component | 0 |
| Isolated Component | 0 |
| Pin Conflict | 2 |
| Breadboard Bypass | 0 |

## Physical Circuit

### Reference



**Success**

| | |
|---|---|
| Redundant Connection | 0 |
| Extraneous Component | 0 |
| Missing Component | 0 |
| Isolated Component | 0 |
| Pin Conflict | 0 |
| Breadboard Bypass | 0 |